
Summary

Optimal land use planning today often requires balancing community values with business profits. Models must take into account factors such as geography, climate, available business options, and community needs and local culture in making important decisions. Now, community leaders and business planners face the challenge of determining the best use for a 3 square kilometer parcel of land for development by determining the optimal combination of buildings to maximize benefits from different perspectives.

We constructed a **map clustering segmentation model** based on information provided by decision makers. First, we designed evaluation indicators for map clustering and scored the map accordingly. The heat map is created afterwards. We then created a binary heat map by setting the threshold value and using **dilation and erosion** to eliminate noise. Finally, we overlapped the binary heat maps of all indicators and performed the final segmentation according to the color shades.

Next, we designed a **multi-factor aggregation evaluation model** by aggregating several linear models with economic, environmental, and community contribution values. We then utilized **AHP** to assign weights to the model.

Lastly, we utilized a **genetic algorithm** to solve the multi-factor aggregation evaluation model, which can be viewed as a linear optimization problem. This involved continuously eliminating lower evaluated plans and mutating and crossing over higher evaluated plans to obtain a more optimal solution. We then solved the problem for the original case and the case after building the fab separately, resulting in their respective optimal solutions.

Our analysis showed that the optimal solution resulted in the greatest benefits in revenue generation, environmental sustainability, and community impact.

In addition to constructing three distinct models, we conducted a **generalization study** to evaluate the effectiveness of our pipeline for all types of land planning problems. We also performed **sensitivity and robustness analysis** on both the evaluation model and genetic algorithm. This verified the strength of our model while identifying variables with significant impact on the results.

Our research is highlighted by the **complete pipeline** we presented, the utilization of **CV's** map clustering segmentation model to significantly simplify the task and reduce information loss, and the genetic algorithm which assisted in rapidly finding the relative optimal solution from a large search space. Hence, the benefits of our research include **strong generalizability** and inclusion of **multiple aspects** for consideration. However, the lack of abundant high-quality data and high computational resource requirements impose limitations to some extent.

Keywords: Map clustering, Map segmentation, Multi-factor evaluation, AHP, Genetic Algorithm

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1 Introduction

1.1 Background

Land optimization plays a crucial role in resource management as we face a growing demand for land resources and the need for sustainable development. In order to optimize land use, it is necessary to carefully consider a variety of factors such as the geological location, climate, elevation, slope, tree cover, and land cover, as well as the economic and social implications of any land use decision. Making informed decisions about land use often requires decision-makers to confront complex and conflicting trade-offs. In these situations, constructing effective and robust mathematical models can be essential for guiding decision-making in a complex and dynamic environment. By leveraging advanced computational techniques such as computer vision and genetic evolution algorithms, these models can help decision-makers identify optimal land use plans that balance economic, environmental, and community considerations.

Mathematical modelling is a powerful tool that offers a systematic and data-driven approach to optimize land use. It involves capturing the relationships among various variables and simulating the behaviour of the designated parcel of land based on mathematical equations and algorithms. This approach allows for a quantitative assessment of different land use scenarios while considering short-term and long-term economic benefits without subjectivity and bias in decision-making. Mathematical models are also capable of incorporating external elements, such as job creation or environmental impacts, providing a comprehensive analysis of the potential benefits and drawbacks of various choices. These features make mathematical modelling a valuable tool for community leaders and business planners who require a thorough analysis of the potential implications of their decisions.

1.2 Problem Restatement

- The first task requires us to create a quantitative model that takes into account both short and long term benefits, and to create our own evaluation metrics for finding and evaluating reasonable land uses
- The second task requires us to rate and briefly analyze at least two of the land use possibilities we find. A sensitivity analysis was also needed on the various indicators.
- The third task requires considering the impact of the new semiconductor manufacturing plant being built in the nearby town on our model and finding and evaluating alternative land-use options.
- The fourth task requires us to verify the scalability and generalizability of the model.

1.3 Assumptions

1. The parcel of land is available for development and can be used for commercial purposes.
Explanation: Google Earth shows that this area is currently completely undeveloped, so we have determined that there is no undevelopable part

2. The results of the model are based on the assumptions and data inputs used and may not reflect the actual outcomes of the land use planning decisions.

Explanation: Since the data itself needs to be collected by ourselves and there is no absolute optimal solution, the solution we are looking for is a relative optimal solution.

3. The data used to develop the evaluation model is up-to-date and accurate.

Explanation: Since there is no fixed data source, it is impossible to judge the reliability of the data, and we need to assume the accuracy and timeliness of our data

1.4 Our Work

The pipeline created in our work involves several stages to solve the problem of optimal land use planning. First, we applied computer vision techniques such as binary heat map, erosion and dilation to segment the map into different areas. Next, a multi-factor evaluation model was used to evaluate the benefits of different combinations of buildings, taking into account factors such as economic, environmental, and community contributions. Finally, a genetic evolution algorithm was applied to identify the optimal combination of buildings. The output of this pipeline is an optimized land use plan that maximizes the benefits of the chosen combination of buildings.

This approach has several advantages, including the ability to quickly and efficiently evaluate a large number of possible land use plans, and the ability to consider multiple factors in the decision-making process. However, the approach also has some limitations, including the need for accurate data inputs, the potential for bias in the evaluation criteria, and the limitations of the genetic evolution algorithm in finding the true optimal solution. Overall, our work provides a useful framework for addressing the complex problem of optimal land use planning, and can be applied in a variety of different contexts.

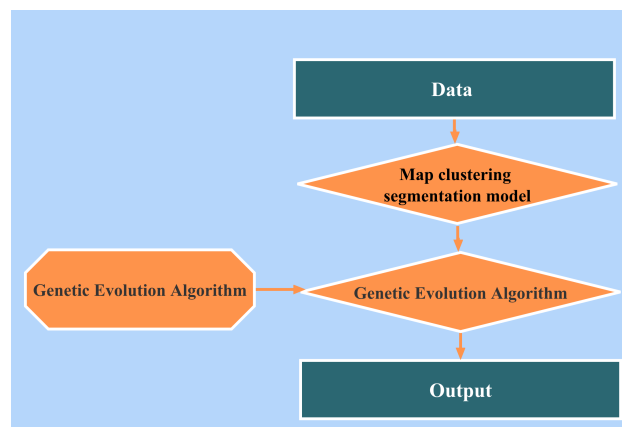


Figure 1: The pipeline of our work

2 Notations

Symbol	Meaning
ϵ_{eco}	economic benefit
ϵ_{env}	environment impact
ϵ_{comm}	community contribution
C_0	the initial investment
C_{short}	expected inflow in short term
C_{long}	expected inflow in short term
r	discount rate
t	time period
$\beta_{construct}$	the unit construction cost
δ	surface
ζ	marketing cost
α_1	unit benefit for short term
α_2	unit benefit for long term
β_{labor}	unit labor cost
$\beta_{hardware}$	unit hardware maintenance cost
μ_{soil}	negative impact on soil quality
μ_{water}	negative impact on water quality
μ_{bio}	negative impact on bio quality
μ_{gas}	negative impact on gas quality
θ_{employ}	employment opportunities provided
θ_{volun}	volunteering opportunities provided

3 Map Clustering Segmentation Model (coarse planning)

Based on the multiple given maps with several different classifications, we will try to quantify the target land into a finite number of cells and form a mesh structure using a grid clustering method to roughly divide the areas with different characteristics.

3.1 Design evaluation indicators

We have collected most of the factors that may affect land use:

Based on the resources that can be given in the question and the data we have collected we have selected these evaluation indicators to score the land for zoning: **Transportation, Slope Angle, Signal Coverage, Soil Fertility, Water Supply and Vegetation Coverage**

3.2 Evaluation Criteria

1.Traffic is set up to evaluate indicators 1-4 depending on the presence or absence of roads around this area.Indicator 4 has the most convenient access.

2.Slope Angle is evaluated by indicators 1-6, which increase in order according to whether the area is crossed by contour lines and whether the contour lines are dense or thickened.Indicator 6 has the steepest slope.

3.Signal Coveragehas only two evaluation indicators yes or no, 0 and 1.

4. Soil Fertility is set up with 5 evaluation indicators. The land is judged to have the best soil quality by whether it has been reclaimed for cultivation and the steepness of the slope from 1 to 5. 5 is judged to have the best soil quality.

5. Water Supply has three criteria depending on whether the land has water flowing through it or not: 0 is no water source around or on the land itself, 1 is a water source passing nearby, and 2 is a water source passing through the land.

6. Vegetation Coverage has five evaluation indicators for vegetation cover. From 1 to 5, they depend on the height and density of vegetation on the land. 1 is all cleared for farmland, 5 is patches of tall forest

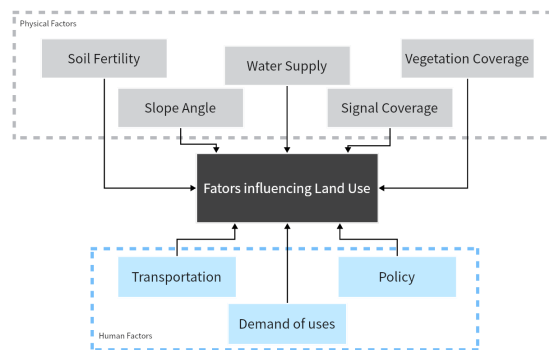


Figure 2: All factors on land use

3.3 Heat Map

We use heat maps to express the scoring of different evaluation metrics for the same plot of land:

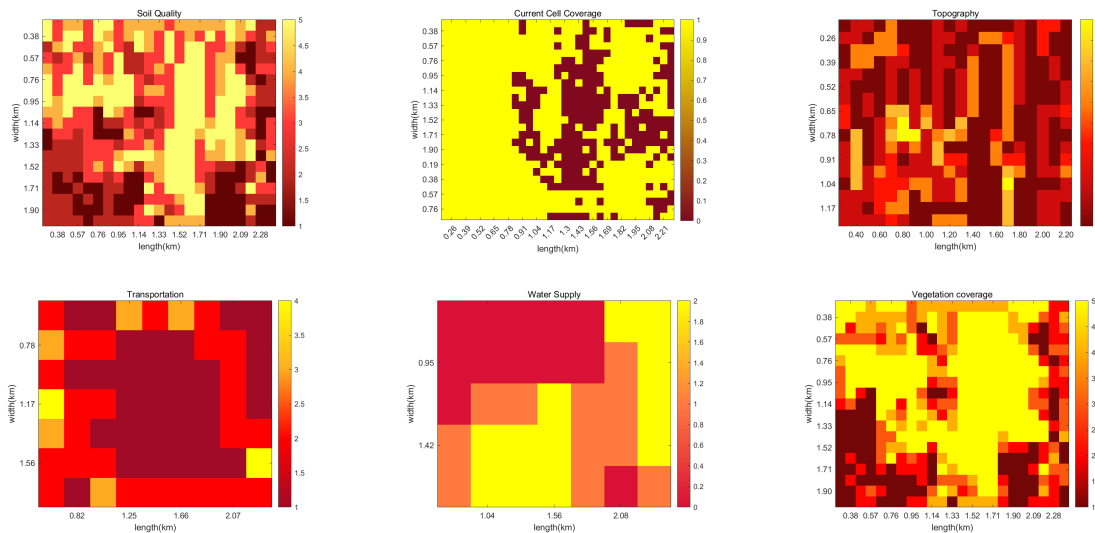


Figure 3: Heat Map of six indicators

3.4 Image Processing

In order to divide the image of the map more convenient and intuitive, we introduced the methods and concepts of Image Binarization and Binary Dilation and Erosion.

3.4.1 Image Binarization

We first use the thresholding method for binarization to segment the image. Again a fixed threshold is set to classify multiple evaluation metrics. We choose the middle number of each evaluation criterion as the threshold, and images larger than this threshold are set to 1, and those smaller than this threshold are set to 0, i.e., all the criteria are classified into two categories.

3.4.2 Binary Dilation and Erosion

Usually we cycle a combination of corrosion and expansion, with expansion filling in the gaps and corrosion eliminating the noise.

$$Dilation : A \oplus B = \{Z | (\hat{B})_z \cap A \neq \emptyset\} \quad (1)$$

$$Erosion : A \ominus B = \{Z | (B)_z \subseteq A\} \quad (2)$$

3.4.3 Application of binarized heat map

According to the above method we have manipulated the image of the given land with different factors:

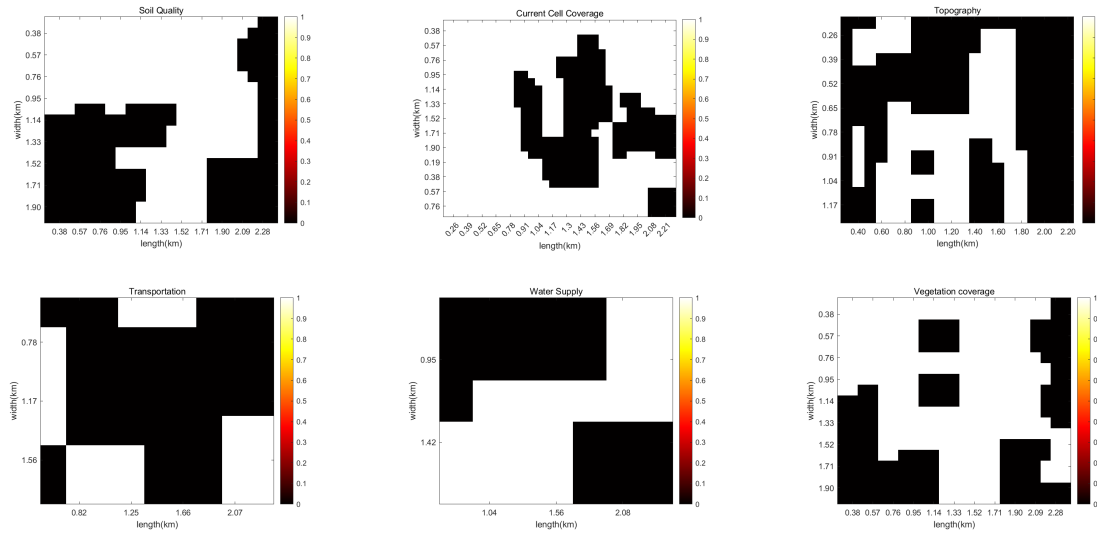


Figure 4: Clustering map after image processing

3.5 Map Segmentation Results

To visually delineate the different clusters, we clustered the binarized heat maps of six different factors by reducing the transparency and clustering them together

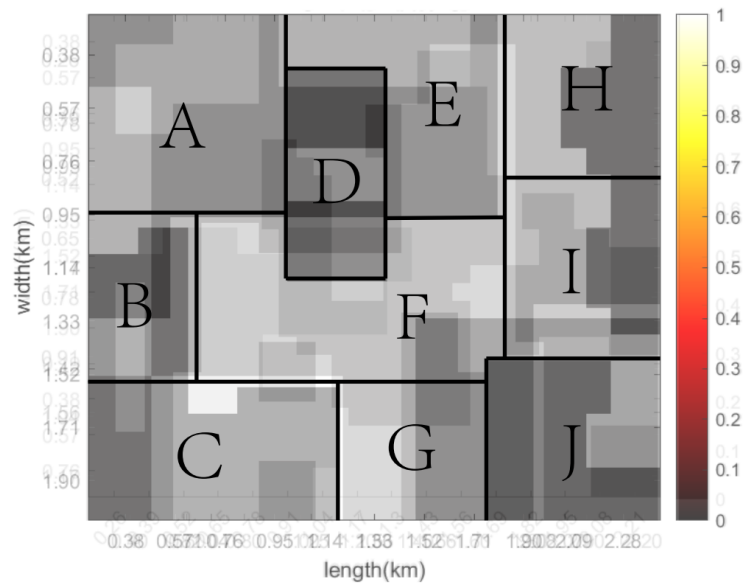


Figure 5: Clustering results from multiple factors

Overview of each region		
Region	Brief introduction	Area
A	Scarcity of water, high signal coverage, high quality soil, high vegetation cover, undulating slopes, low accessibility	$0.84km^2$
B	Enough water resources, high signal coverage, low soil quality, low vegetation cover, undulating slopes, easy access	$0.21km^2$
C	Abundant water resources, high signal coverage, very low soil quality, low vegetation cover, extremely undulating slopes, easy access	$0.66km^2$
D	Scarcity of water, low signal coverage, high soil quality, low vegetation cover, undulating slopes, low accessibility	$0.30km^2$
E	Scarcity of water, low signal coverage, high soil quality, high vegetation cover, gentle slope, moderate accessibility	$0.61km^2$
F	Adequate water resources, low signal coverage, loose soil, high vegetation cover, extremely steep slopes, extremely isolated traffic	$0.55km^2$
G	Adequate water resources, high signal coverage, high soil quality, high vegetation cover, extremely steep slopes, extremely isolated traffic	$0.38km^2$
H	Adequate water resources, high signal coverage, low soil quality, low vegetation cover, gentle slopes, and extremely closed access	$0.36km^3$
I	Sufficient water resources, low signal coverage, low soil quality, low vegetation cover, gentle slope, low accessibility	$0.47km^2$
J	Scarcity of water resources, low signal coverage, low soil quality, low vegetation cover, gentle slope, convenient transportation	$0.62km^2$

4 Multi-factor Aggregation Evaluation Model (fine planning)

After coarse planning using the map clustering segmentation model in the first part, we succeeded in reducing the complexity of the problem.

In this part, we will build a multi-factor aggregation evaluation model in terms of **economic benefits, environmental impacts, and community contributions** to achieve the goal of considering land planning using a quantitative approach. We will build this evaluation model by multi-module aggregation, so we will first show the big model with target as the goal, and then show each sub-model separately.

4.1 Target

We found that if land planning is considered only from the economic point of view, then with the goal of maximizing profits, problems such as serious damage to the environment and community dissatisfaction with land planning are likely to occur. First, it will affect the reputation of the builder, subsequent development, etc., and second, it will affect the long-term benefits of the project. Based on these understandings, we constructed the following quantitative evaluation model based on the objective of strong generalizability:

$$target = w_1 \times \epsilon_{eco} + w_2 \times \epsilon_{env} + w_3 \times \epsilon_{comm} \quad (3)$$

First we will explain in detail the corresponding part of this model:

1. target: Objective of quantitative evaluation of the value derived from land use
2. ϵ_{eco} : economic benefit, The results of the analysis are quantified from the perspective of economic benefits.
3. ϵ_{env} : environment impact, The results of the analysis are quantified in terms of environmental impacts.
4. ϵ_{comm} : community contribution, The results of the analysis are quantified from the perspective of community contribution.
5. w_i represents the weight of each part respectively.

The targets we construct take into account the issues that arise above, and in addition to economic benefits, we also add two aspects of environmental impact and community contribution value, truly achieving a comprehensive consideration from multiple perspectives. We can adjust our targets in a controlled manner by giving different weights to obtain a more accurate evaluation model. We will complete the construction of the three sub-models of economic benefits, environmental impact and community contribution value in the following three sections.

4.2 Economic

$$\epsilon_{eco} = -C_0 + \Sigma[C_{short}/(1+r)t] + \Sigma[C_{long}/(1+r)t] \quad (4)$$

First we will explain in detail the corresponding part of this model:

1. ϵ_{eco} here stands for economic benefits being evaluated.
2. C_0 represents the initial investment made at the beginning of the project.
3. C_{short} represents the expected cash inflow in each period in short term (in 5 years).
4. C_{long} represents the expected cash inflow in each period in long term (5-20 years).
5. r represents the discount rate or the rate of return that investors require for the project.
6. t represents the time period in which the cash inflows occur.

Therefore, this section generalizes the evaluation criteria of economic benefits by considering the initial cost, short-term cost and long-term cost. After that, we will model more specifically for C_0 , C_{short} and C_{long} so that they can be applied to the problem we are facing.

4.2.1 Initial cost

$$C_0 = \beta_{construct} \times \delta + \zeta \quad (5)$$

1. C_0 represents the initial cost of the construction project.
2. $\beta_{construct}$ represents the unit construction cost, which is the cost of constructing one unit (e.g. one square kilometer) of the building or structure.
3. δ represents the surface area of the building or structure being constructed.
4. ζ represents any marketing costs associated with the construction project.

4.2.2 Short-term Benefits

$$C_{short} = \alpha_1 \times \delta - w_4 \times \beta_{labor} \times \delta - w_5 \times \beta_{hardware} \times \delta - w_6 \times \zeta \quad (6)$$

1. C_{short} represents the short-term benefit of the project or investment being evaluated.
2. α_1 represents the unit benefit. It is the amount of revenue generated by one unit of surface area.
3. δ represents the total surface area of the project being evaluated.
4. β_{labor} represents the unit labor cost. It is the cost of labor required to construct or operate the project, per unit of surface area.
5. $\beta_{hardware}$ represents the unit hardware maintenance cost.
6. ζ represents any marketing costs associated with the project.
7. w_i represents the weight or importance assigned to each factor in the calculation of short-term benefits.

4.2.3 Long-term Benefits

$$C_{long} = \alpha_2 \times \delta - w_7 \times \beta_{labor} \times \delta - w_8 \times \beta_{hardware} \times \delta - w_9 \times \zeta \quad (7)$$

This part is almost the same with the former short-term benefits, only the weight to each factor and the unit benefit vary from the short-term formula, which makes it different from it.

4.3 Environment

$$\epsilon_{env} = w_{10} \times \mu_{soil} + w_{11} \times \mu_{water} + w_{12} \times \mu_{bio} + w_{13} \times \mu_{gas} \quad (8)$$

1. ϵ_{env} represents the overall environmental impact of a project or activity being evaluated.
2. w_i Represents the weight assigned to each of the four factors in the calculation of environmental impact.
3. μ_{soil} represents the negative impact of the project on soil quality.
4. μ_{water} represents the negative impact of the project on water quality.
5. μ_{bio} represents the negative impact of the project on biodiversity.
6. μ_{gas} represents the negative impact of the project on atmospheric quality.

4.4 Community

$$\epsilon_{comm} = w_{14} \times \theta_{employ} + w_{15} \times \theta_{volun} \quad (9)$$

1. ϵ_{comm} represents the overall community impact of a project or activity being evaluated.
2. w_i represents the weight assigned to each of the two factors in the calculation of community impact.
3. θ_{employ} represents the possibility of the project to provide employment opportunities to members of the community.
4. θ_{volun} represents the possibility of the project to provide volunteering opportunities to members of the community.

At this point we have completed the construction of all sub-models of the multi-factor aggregation evaluation model. After aggregation, our model can be shown in the following figure.

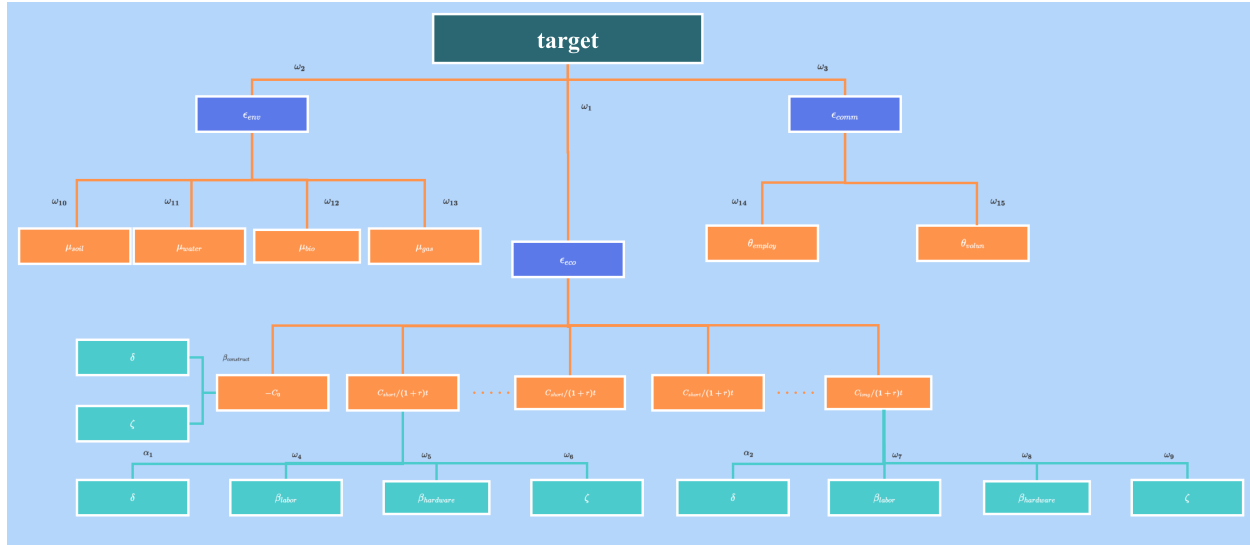


Figure 6: The final Multi-factor aggregation evaluation model

4.5 AHP (The Analytic Hierarchy Process)

Although we have completed the modeling of all evaluation indicators, from the perspective of solving the actual problem, there are still a large number of unknown variables in our model, which will largely affect the complexity of the calculation in the subsequent solution of the optimal case, so we will use hierarchical analysis to determine all weights before the actual solution.

4.5.1 Introduction of AHP

The Analytic Hierarchy Process (AHP) is a structured decision-making technique that helps individuals and organizations make complex decisions involving multiple criteria and alternatives.

The AHP process involves breaking down a complex decision into smaller, more manageable parts. The decision maker creates a hierarchy of criteria and alternatives, where each criterion is a sub-factor of the decision, and each alternative is a potential solution to the decision. The criteria and alternatives are then ranked and compared against each other to determine the best course of action.

One of the unique features of AHP is its ability to accommodate both objective and subjective factors in the decision-making process. We found it pretty useful in our decision making, because the land management contains a lot of subjective factors so that we can make sure there exist ways for us to inference the mode.

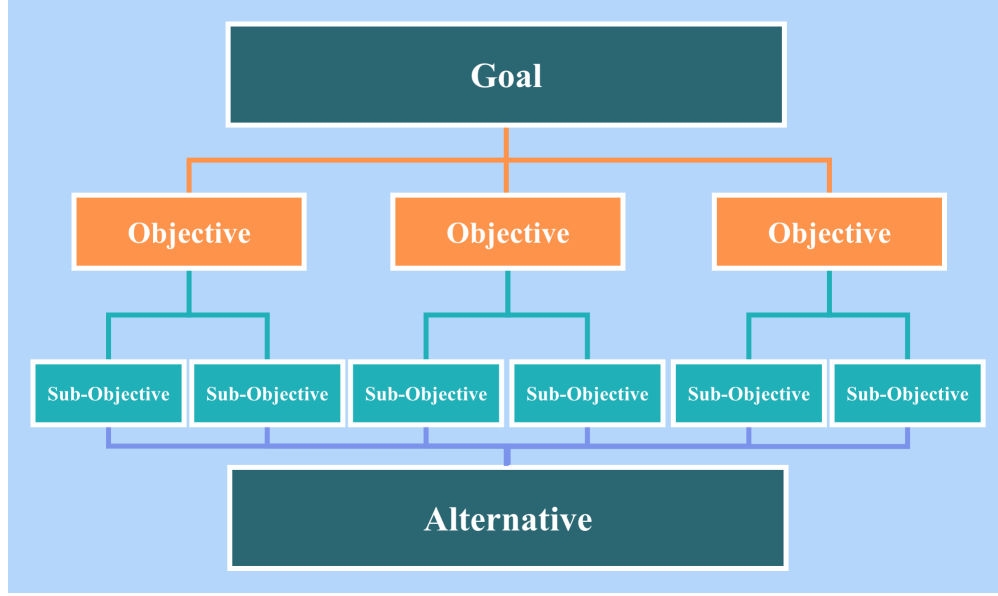


Figure 7: The structure of AHP

4.5.2 Application of Hierarchical Analysis

In this problem, we use 4.5.1 as our decision architecture.

- Objective : $\epsilon_{eco}, \epsilon_{env}, \epsilon_{comm}$
- Sub-Objective:
 - $\epsilon_{eco} : C_0, C_{short}, C_{long}$
 - $\epsilon_{env} : \mu_{soil}, \mu_{water}, \mu_{bio}, \mu_{gas}$
 - $\epsilon_{comm} : \theta_{employ}, \theta_{volun}$

Although we have completed the modeling of all evaluation indicators, from the perspective of solving the actual problem, there are still a large number of unknown variables in our model, which will largely affect the complexity of the calculation in the subsequent solution of the optimal case, so we will use hierarchical analysis to determine all weights before the actual solution.

In contrast to the full decision structure, our decision hierarchy is not specified for now, and we will use genetic algorithms for optimal selection in Section 4.

4.5.3 Weight Determination

After the decision structure is determined, we can **construct the judgment matrix** and **compute the weight vector** and **consistency test** for the hierarchical single ranking.

1. construct the judgment matrix

To create a judgment matrix for the Analytic Hierarchy Process (AHP), we compared each objective against every other objective at the same level of the hierarchy using a pairwise comparison method.

Step 1: Define the Scale, for simplicity, here we choose 1-5 as our comparison metrics.

Step 2: Pairwise Comparison

Step 3: Populate the Judgment Matrix

Following these steps, we accomplished the creation of all judgment matrix for our objective and sub-objective.

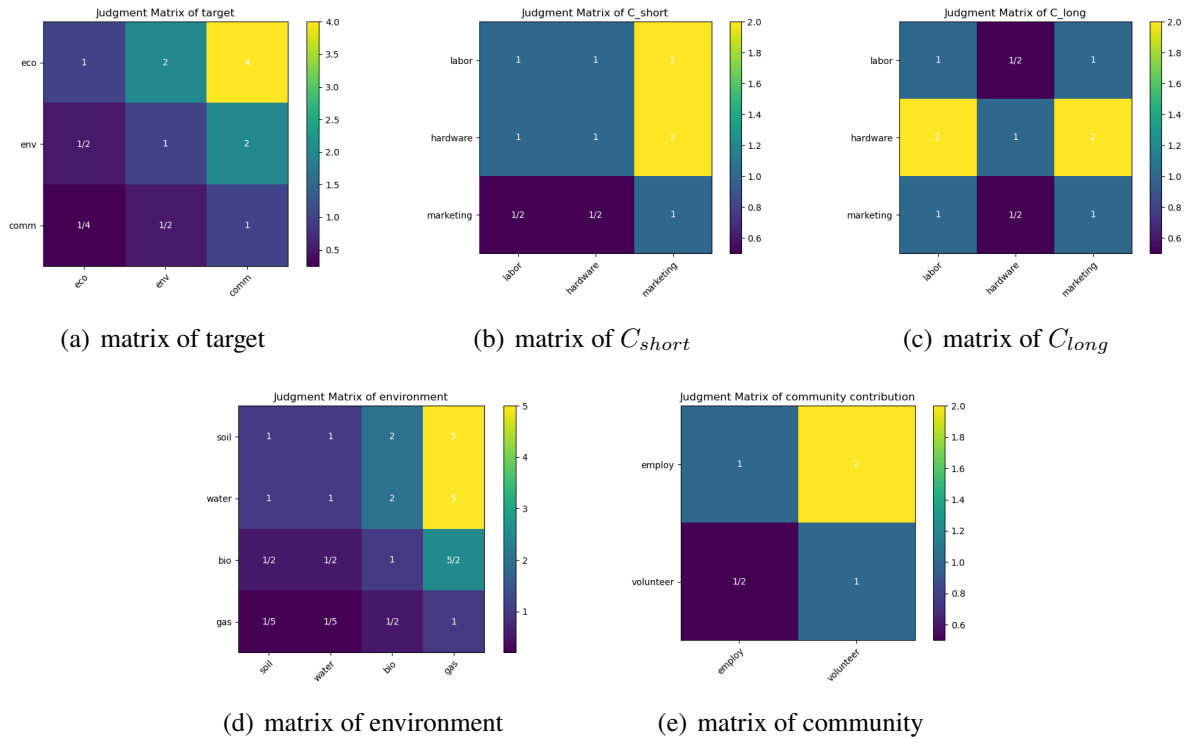


Figure 8: Judgement matrix of our model

2. consistency check

The consistency check is used to ensure that the judgments made in the pairwise comparison matrix are consistent. The consistency check involves comparing the judgments made for each pairwise comparison to see if they are logically consistent with each other.

In AHP, consistency is evaluated using the consistency ratio (CR). The CR is calculated by dividing the consistency index (CI) by a randomly generated value known as the random index (RI), which is a function of the number of criteria being compared. The RI values are provided in a table for different sizes of matrices, and are used to calculate the consistency ratio as follows:

$$CR = CI/RI \quad (10)$$

If the CR value is less than 0.1, the pairwise comparison matrix is considered consistent. If the CR value is greater than 0.1, the pairwise comparison matrix is considered inconsistent. The CI is calculated as follows:

$$CI = (\lambda_{max} - n)/(n - 1) \quad (11)$$

Where λ_{max} is the maximum eigenvalue of the pairwise comparison matrix and n is the number of criteria being compared.

Following steps above, we have checked all the judgement matrix and found them are all consistent.

3. compute the weight vector

The weight vector is used to determine the relative importance of each criterion being evaluated in a decision-making process. The weight vector is generated by combining the judgments made in the pairwise comparison matrix for each criterion and calculating the normalized eigenvector associated with the matrix. This is done by dividing each element of the eigenvector by the sum of all the elements in the eigenvector.

Here's all the weights we get.

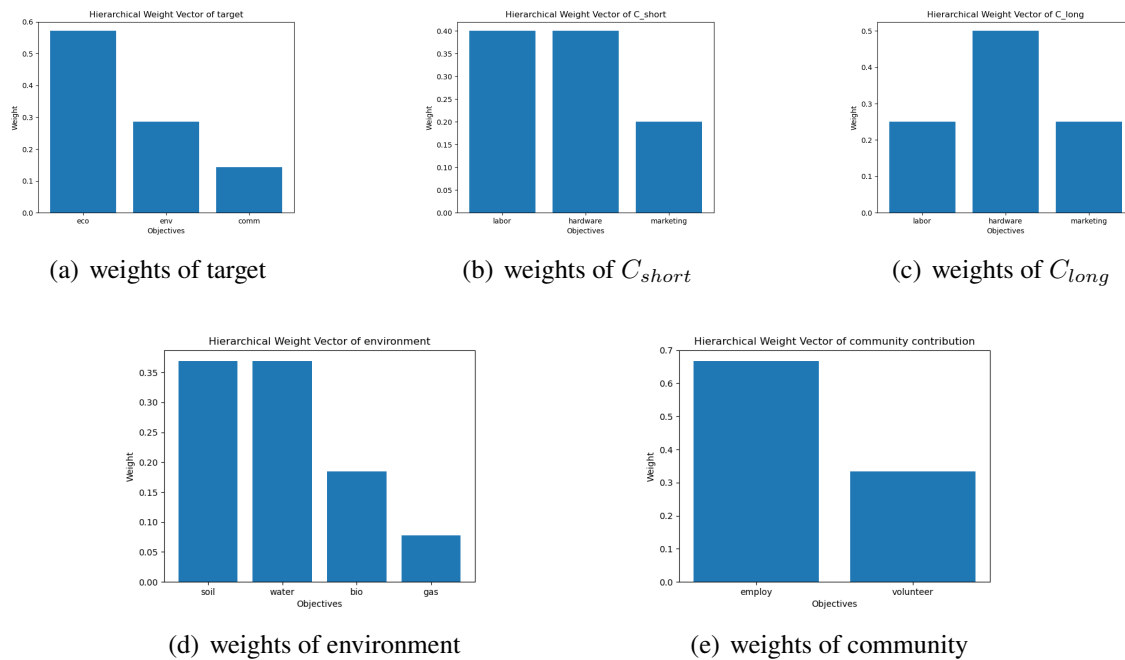


Figure 9: Weights of our model

5 Genetic Algorithm

5.1 Introduction to Genetic Algorithm

Genetic algorithms are a powerful optimization technique inspired by the process of natural selection and genetics. They are often used in mathematical modeling problems where the goal is to find the optimal values of a set of parameters.

In a genetic algorithm, a population of potential solutions is created and evolved over time through the use of genetic operators such as crossover, mutation, and selection. Each potential solution in the population is represented by a string of values, called a chromosome or genotype, and is evaluated based on its fitness, which is a measure of how well it solves the mathematical problem at hand.

The genetic algorithm applies these genetic operators to create new generations of potential solutions until a stopping criterion is met. The best solution found during the evolution process is typically the chromosome with the highest fitness value.

Genetic algorithms are particularly useful for solving complex optimization problems with a large search space, which fits our problem well, so we believe using this algorithm could help us finding the optimized solution.

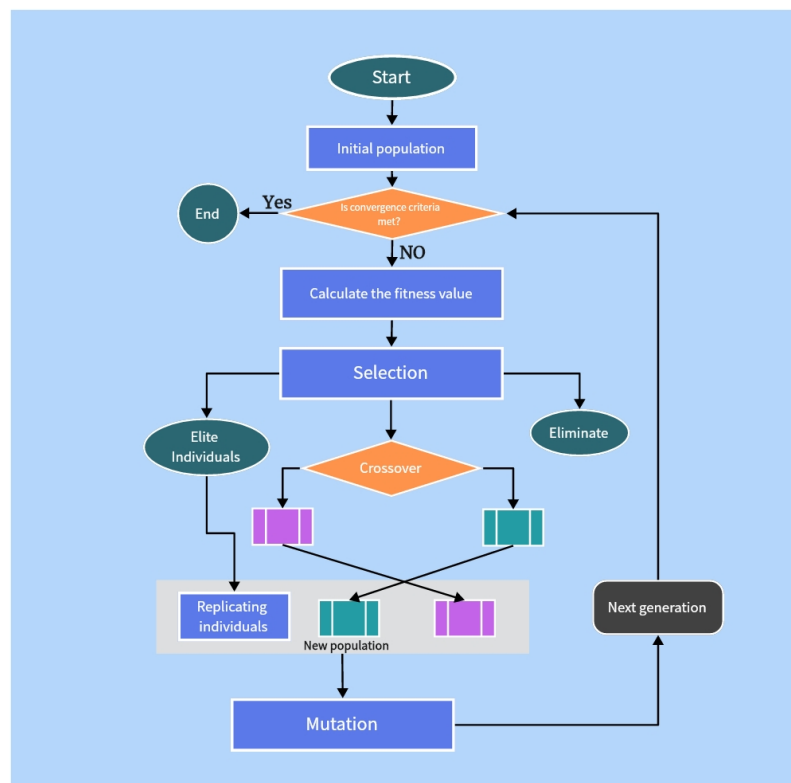


Figure 10: Flow chart of Genetic Algorithm

5.2 Application of Genetic Algorithm

The migration of the genetic algorithm to our problem requires clarification of what the following definitions represent in our problem:

- Individual: one building unit to be constructed in each region of 3.5, one individual for each combination of permutations.
- Population: the set of all possible combinations of permutations, containing the set of all possibilities for building unit construction according to the chunks of 3.5.
- Fitness function: the evaluation function used to determine whether a certain individual is an elite individual, here we use the multi-factor aggregation evaluation model built in the previous section as the fitness function.
- Crossover: Occurs between elite individuals, i.e., the building units of a certain area of an individual with a higher fitness function score are exchanged with another one.
- Mutation: Occurs in a single elite individual, i.e., the building units of some regions of an individual with a high fitness function score are randomly changed.

After completing these defined meanings, since the parameters used by the fitness function for different individuals, such as $\beta_{construct}$, β_{labor} , $\beta_{hardware}$, μ_{soil} , μ_{water} , μ_{bio} , μ_{gas} , θ_{employ} , θ_{volun} , r , etc. are different, and if we blindly use the same parameters will lead to the adaptation function of the genetic algorithm cannot use the map information reasonably, so we need to give the value of each parameter for different buildings in different areas to ensure that the calculation of the adaptation function is accurate. After a lot of searching and summarizing, we have selected values for each parameter for different buildings at different areas to make it fit the actual situation. Here we have selected some parameters to show 5.2, and we cannot show all of them here because the total number of parameters reaches 1120 (7(building units) $\times$ 10(possible areas) $\times$ 16(parameters)).

At this point we have completed all the preconditions for using the genetic algorithm, and we will use the procedure outlined in the following algorithm 1 to select the top 10 optimal solutions.

		cross-									cross-								
		country	crop	grazing	regenerative	solar	agrivoltaic	agritourist			country	crop	grazing	regenerative	solar	agrivoltaic	agritourist		
		skiing	farm	farm/ranch	farm	array	farm	center			skiing	farm	farm/ranch	farm	array	farm	center		
		facility									facility								
β_{labor}	A	149,287	99524.667	124405.8333	82937.22222	82937.2	82937.22222	149287			A	-0.87	-0.3915	-0.5481	-0.340605	-0.8265	-0.619875	-0.657068	
	B	125,923	83948.667	104935.8333	69957.22222	69957.2	69957.22222	125923			B	-0.25	-0.1125	-0.1575	-0.097875	-0.2375	-0.178125	-0.188813	
	C	125,923	83948.667	104935.8333	69957.22222	69957.2	69957.22222	125923			C	-0.17	-0.0765	-0.1071	-0.066555	-0.1615	-0.121125	-0.128393	
	D	155,298	103532	129415	86276.66667	86276.7	86276.66667	155298			D	-0.54	-0.243	-0.3402	-0.21141	-0.513	-0.38475	-0.407835	
	E	140,192	93461.333	116826.6667	77884.44444	77884.4	77884.44444	140192			E	-0.87	-0.3915	-0.5481	-0.340605	-0.8265	-0.619875	-0.657068	
	F	180,062	120041.33	150051.6667	100034.4444	100034	100034.4444	180062			F	-0.66	-0.297	-0.4158	-0.25839	-0.627	-0.47025	-0.498465	
	G	155,298	103532	129415	86276.66667	86276.7	86276.66667	155298			G	-0.87	-0.3915	-0.5481	-0.340605	-0.8265	-0.619875	-0.657068	
	H	112,550	75033.333	93791.66667	62527.77778	62527.8	62527.77778	112550			H	-0.17	-0.0765	-0.1071	-0.066555	-0.1615	-0.121125	-0.128393	
	I	125,923	83948.667	104935.8333	69957.22222	69957.2	69957.22222	125923			I	-0.17	-0.0765	-0.1071	-0.066555	-0.1615	-0.121125	-0.128393	
	J	125,923	83948.667	104935.8333	69957.22222	69957.2	69957.22222	125923			J	-0.17	-0.0765	-0.1071	-0.066555	-0.1615	-0.121125	-0.128393	

(a) The value of each possible permutation of β_{labor} (b) The value of each possible permutation of μ_{soil}

		cross-								
		country	crop	grazing	regenerative	solar	agrivoltaic	agritourist		
		skiing	farm	farm/ranch	farm	array	farm	center		
		facility								
α_1	A	746,435	796197.33	995246.6667	746435	829372	663497.7778	895722		
	B	629,615	671589.33	839486.6667	629615	699572	559657.7778	755538		
	C	629,615	671589.33	839486.6667	629615	699572	559657.7778	755538		
	D	776,490	828256	1035320	776490	862767	690213.3333	931788		
	E	700,960	747690.67	934613.3333	700960	778844	623075.5556	841152		
	F	900,310	960330.67	1200413.333	900310	1000344	800275.5556	1080372		
	G	776,490	828256	1035320	776490	862767	690213.3333	931788		
	H	562,750	600266.67	750333.3333	562750	625278	500222.2222	675300		
	I	629,615	671589.33	839486.6667	629615	699572	559657.7778	755538		
	J	629,615	671589.33	839486.6667	629615	699572	559657.7778	755538		

(c) The value of each possible permutation of α_1

Figure 11: Possible values of some parameters

Algorithm 1 Genetic Evolution

```

 $t \leftarrow 0$                                 ▷ initialize time step
initialize( $P(t = 0)$ )                      ▷ initialize population
evaluate( $P(t = 0)$ )                          ▷ evaluate initial population
while isNotTerminated( $P(t)$ ) do          ▷ while not reaching seuil value of the fitness function
    for all  $p$  in  $P(t)$  do
        evaluate( $p$ )                        ▷ evaluate individuals
    end for
     $P(t + 1) \leftarrow$  choosingElite( $P$ )      ▷ reproduce the next generation
    mutate( $P(t + 1)$ )                        ▷ do the mutation operation for  $P(t + 1)$ 
    crossover( $P(t + 1)$ )                    ▷ do the crossover operation for  $P(t + 1)$ 
     $t \leftarrow t + 1$ 
end while
return  $P(t)$ 

```

Through extensive calculations we obtained 10 optimal solutions, of which we have selected two more representative ones as examples to explain:

We can find that because our chunks are more fragmented, so there is a situation where the optimal solution will split a building into more than one, at this time we will use manual screening to sieve out and select the optimal solution of the same building plan in the adjacent area as the final solution.

Therefore, for this problem, we give the optimal solution as 12(b), and the score of the fitness function is **2833942.223**. As for 12(a), the score is **2827930.461**, which is quite similar to the 12(b), but due to the land plan is too fragmented.

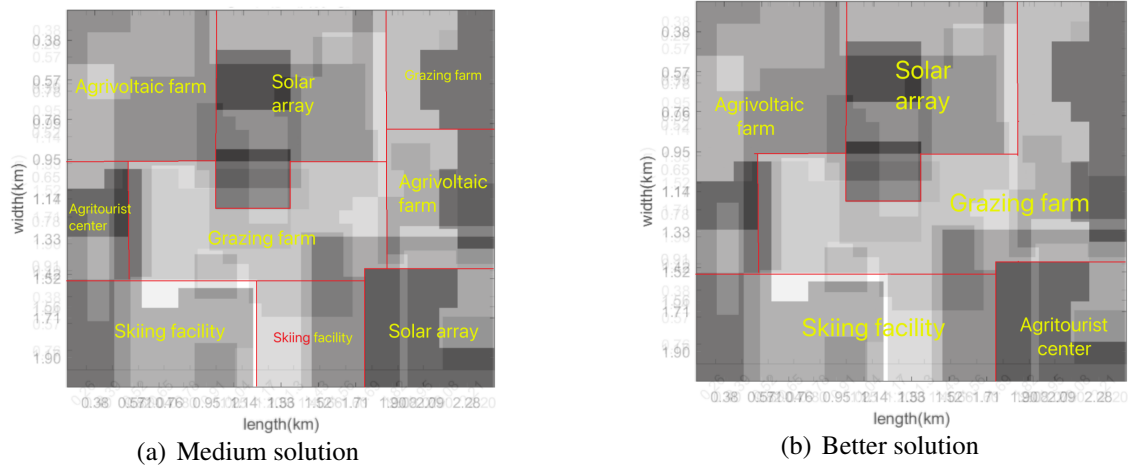


Figure 12: Comparison of 2 optimized solution

5.3 Special Case

The given problem involves the construction of a new fab that would generate numerous job opportunities. With the construction of the new facility, human activities in the surrounding area are expected to increase, leading to a surge in demand for various resources, such as agricultural and sideline food, electricity, and entertainment, thereby increasing the overall revenue. However, it is important to consider the flip side of the scenario. The increase in job opportunities may reduce the attractiveness of the employment options provided and increase the labor costs due to the increased competition. Furthermore, the rise in human traffic may also result in environmental pollution. To analyze the impact of these factors, we have modified the results of the table, corresponding to the aforementioned aspects.

Also based on 1, after selecting the optimal first 10 solutions, the optimal solutions are manually filtered.

		cross-country skiing facility								cross-country skiing facility							
		crop farm	grazing farm/ranch	regenerative farm	solar array	agrivoltaic farm	agritourist center			crop farm	grazing farm/ranch	regenerative farm	solar array	agrivoltaic farm	agritourist center		
β_{labor}	A	194,073	129382.0667	161727.5833	107818.3889	107818.3889	107818.3889	194073.1	μ_{soil}	A	-1.131	-0.50895	-0.71253	-0.4427865	-1.07445	-0.8058375	-0.85418775
	B	163,700	109133.2667	136416.5833	90944.38889	90944.38889	90944.38889	163699.9		B	-0.325	-0.14625	-0.20475	-0.1272375	-0.30875	-0.2315625	-0.24545625
	C	163,700	109133.2667	136416.5833	90944.38889	90944.38889	90944.38889	163699.9		C	-0.221	-0.09945	-0.13923	-0.0865215	-0.20995	-0.1574625	-0.16691025
	D	201,887	134591.6	168239.5	112159.6667	112159.6667	112159.6667	201887.4		D	-0.702	-0.3159	-0.44226	-0.274833	-0.6669	-0.500175	-0.5301855
	E	182,250	121499.7333	151874.6667	101249.7778	101249.7778	101249.7778	182249.6		E	-1.131	-0.50895	-0.71253	-0.4427865	-1.07445	-0.8058375	-0.85418775
	F	234,081	156053.7333	195067.1667	130044.7778	130044.7778	130044.7778	234080.6		F	-0.858	-0.3861	-0.54054	-0.335907	-0.8151	-0.611325	-0.6480045
	G	201,887	134591.6	168239.5	112159.6667	112159.6667	112159.6667	201887.4		G	-1.131	-0.50895	-0.71253	-0.4427865	-1.07445	-0.8058375	-0.85418775
	H	146,315	97543.33333	121929.1667	81286.11111	81286.11111	81286.11111	146315		H	-0.221	-0.09945	-0.13923	-0.0865215	-0.20995	-0.1574625	-0.16691025
	I	163,700	109133.2667	136416.5833	90944.38889	90944.38889	90944.38889	163699.9		I	-0.221	-0.09945	-0.13923	-0.0865215	-0.20995	-0.1574625	-0.16691025
	J	163,700	109133.2667	136416.5833	90944.38889	90944.38889	90944.38889	163699.9		J	-0.221	-0.09945	-0.13923	-0.0865215	-0.20995	-0.1574625	-0.16691025

(a) value of each possible permutation of β_{labor} (b) New value of each possible permutation of μ_{soil}

	cross-country skiing facility	crop farm						
		grazing farm/ranch	regenerative farm	solar array	agrivoltaic farm	agritourist center		
α_1	A	1,261,475	1345573.493	1681966.867	1261475.15	1401639.056	1121311.244	1513770.18
	B	1,064,049	1134985.973	1418732.467	1064049.35	1182277.056	945821.6444	1276859.22
	C	1,064,049	1134985.973	1418732.467	1064049.35	1182277.056	945821.6444	1276859.22
	D	1,312,268	1399752.64	1749690.8	1312268.1	1458075.667	1166460.533	1574721.72
	E	1,184,622	1263597.227	1579496.533	1184622.4	1316247.111	1052997.689	1421546.88
	F	1,521,524	1622958.827	2028698.533	1521523.9	1690582.111	1352465.689	1825828.68
	G	1,312,268	1399752.64	1749690.8	1312268.1	1458075.667	1166460.533	1574721.72
	H	951,048	1014450.667	1268063.333	951047.5	1056719.444	845375.5556	1141257
	I	1,064,049	1134985.973	1418732.467	1064049.35	1182277.056	945821.6444	1276859.22
	J	1,064,049	1134985.973	1418732.467	1064049.35	1182277.056	945821.6444	1276859.22

(c) New value of each possible permutation of α_1

Figure 13: New possible values of some parameters

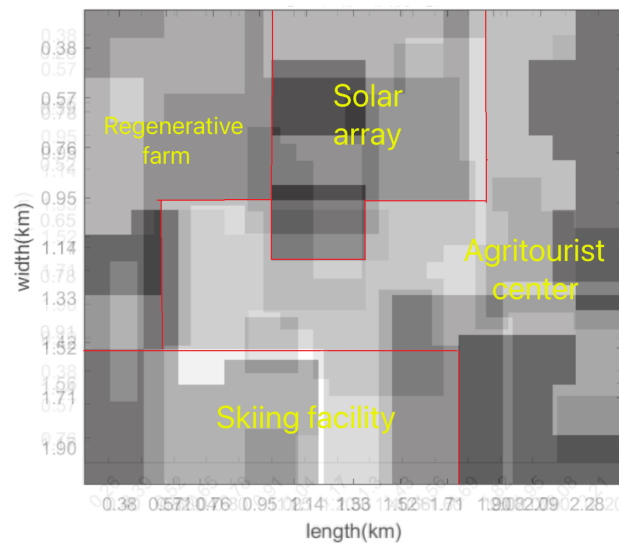


Figure 14: Optimized solution in special case

6 Sensitivity and Robustness Analysis

In addition, the sensitivity and robustness of the model have been carefully studied. The study here focuses on the multi-factor aggregation evaluation model and the genetic algorithm.

6.1 Analysis of Multi-factor Aggregation Evaluation model

In order to test the stability of the model and the importance of the parameters, we applied a certain degree of random correction ($\pm 5\%$) to each of all parameters and used the genetic algorithm to find the optimal solution. After completing the test for all parameters, it can be found that small fluctuations in most parameters do not have a huge impact on the final results, and the overall layout arrangement is still not much different from that before the adjustment, but the final results provided by the genetic algorithm after the modification of the economic part including β_{labor} , $\beta_{hardware}$, α_1 , α_2 and ζ are significantly different from the original results, so we judge that these parameters have a more important influence in the multi-factor aggregation evaluation model.

6.2 Analysis of Genetic Algorithm

As for the robustness of genetic algorithm, we mainly tested different initial population size (1000,2000,5000), the mutation rate(10%,20%,30%) and the crossover rate(10%,20%,30%). We found the latter 2 parameters have a minor impact, but with different initial population size, the solution differs. Due to the ability of the genetic algorithm depends on the computing ability, we could only try to reach a balance between computing time and the quality of the solution.

7 Generalizability Analysis

7.1 Spatial Generalizability

Our model's spatial generalizability refers to its ability to be applied to different geographic areas with different characteristics (e.g., urban, suburban, rural). The extent to which our model can be generalized spatially depends on the availability and quality of data and the similarity of the underlying factors and relationships across different areas. We believe the factors we have decided to use could be applied to most of the cases.

7.2 Temporal Generalizability

Our model's temporal generalizability refers to its ability to be applied over time, accounting for changes in environmental, social, and economic factors. Our model's generalizability over time depends on the adaptability of the factors and evaluation criteria, as well as the ability to incorporate changing data inputs. And we are pretty confident that with the using of 4 which could evaluate the decision in a long range of time period offers us a huge temporal generalizability.

7.3 Cultural and Social Generalizability

Our model's cultural and social generalizability refers to its ability to account for cultural and social differences across different communities. The generalizability of our model in this regard depends on the factors used for evaluation and the extent to which they account for cultural and social differences, like the counting of environmental impact and community contribution and so on.

8 Evaluation of Strengths and Weaknesses

1. Strength

- Incorporates a multi-factor evaluation model: By evaluating multiple factors (e.g., economic, environmental, community contributions) simultaneously, our model can account for the complexity and interdependence of various aspects of land use. This can help ensure that the land use plan is balanced and sustainable.
- Uses computer vision techniques: By leveraging computer vision, our model can automate the process of segmenting the map into different areas, which can save time and reduce errors compared to manual segmentation.
- Utilizes genetic evolution algorithm for optimization: Genetic algorithms can help identify optimal solutions in complex and non-linear optimization problems, such as land use planning. This can help ensure that the plan is optimized for the desired outcomes.

2. Weakness

- Assumes fixed factors and weights: The factors and weights used in the evaluation model are assumed to be fixed and predetermined. However, these may not accurately capture the preferences or priorities of all stakeholders or may need to be adjusted over time. This could lead to sub-optimal outcomes or require frequent updates to the model.
- Limited by available data: Our model's accuracy is limited by the quality and availability of data inputs. For example, if data is missing or incomplete, the model may not provide an accurate analysis. Additionally, the accuracy of the genetic algorithm may be limited by the size and diversity of the initial population.
- Limited by the computing ability of the hardware: Since we are using genetic algorithm which requires a huge amount of computing ability, if we could not reach the desired condition, the computing of the optimized solution may be time-wasting and inaccurate.

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Dear Community Leaders and Business Planners,

I am writing to provide you with my recommendation for the optimal land use plan based on my recent work, which considers the use of the following buildings: an outdoor sports complex, cross-country skiing facility (3-month season), a crop farm, a grazing farm/ranch, a regenerative farm, a solar array, an agrivoltaic farm, and an agritourist center.

After conducting a thorough analysis of the available data, I have developed a pipeline that involves using computer vision techniques to segment the map into different areas, developing a multi-factor evaluation model to evaluate different combinations of buildings, and using a genetic evolution algorithm to identify the optimal combination of buildings for each area.

Based on our analysis, the optimal combination of buildings for the specific sites identified in the map includes a solar array, an agrivoltaic farm, a grazing farm, a cross-country skiing facility (3-month season), and an agritourist center, which you could find in our paper. This combination of buildings provides a range of benefits, including clean energy production, sustainable agriculture practices, local food production, outdoor recreational opportunities, and community engagement through agritourism.

This land use plan is based on a rigorous evaluation of the available data and is designed to maximize the benefits of different combinations of buildings from the perspective of economic, environmental, and community factors. By prioritizing community contributions, economic benefits, and environmental impact, we can create a land use plan that benefits the community as a whole while also promoting sustainable development.

I believe that this land use plan offers the best balance of economic, environmental, and community benefits, and I sincerely recommend you to consider it carefully as you make your decisions. I am available to discuss my recommendations in more detail and to answer any questions that you may have.

Thank you for your attention and consideration.

Sincerely,
US-12850

